

Optimization

1.2 Applications of optimization: where it is used — and what they have in common

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The same question everywhere

In each case that follows there are three things to identify:

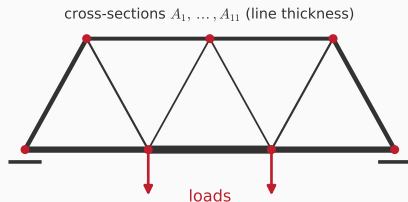
Decision the **decision variables** $\mathbf{x}$ — what is in our hands.

Objective the **objective function** $f(\mathbf{x})$ — the criterion to say that one choice is better than another.

Constraints what cannot be violated.

The physics, economics or biology behind it changes; **the structure of the problem is always the same**. Once it is recognized, the method is chosen — and that is what the course is about.

Structures — minimum-weight truss



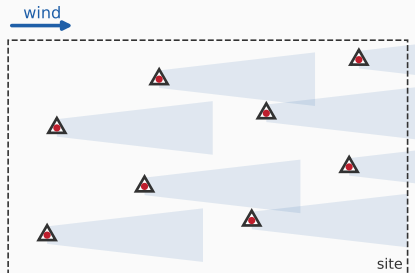
Decision the cross-section A_i of each bar (and, sometimes, the position of the nodes)

Objective the total weight $\sum_i \rho A_i L_i$

Constraints stress in each bar $|\sigma_i| \leq \sigma_{\text{adm}}$;
maximum displacements; buckling;
 $A_i \geq A_{\text{min}}$

Type continuous, nonlinear (the stresses depend on $\mathbf{x}$ through the equilibrium equations), constrained. Ch. 4

Energy — wind farm layout



Decision the coordinates (x_i, y_i) of each turbine

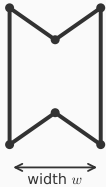
Objective the annual energy produced, minus the wake losses (downwind turbines receive less wind)

Constraints minimum distance between turbines; site boundary; exclusion zones

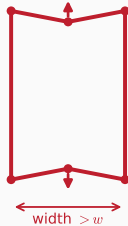
Type continuous, nonlinear, **multimodal** (many local optima) — motivates the global optimization methods of Ch. 5. Ongoing research at IDMEC (M4WIND project)

Materials — auxetic structures

re-entrant cell (at rest)



in tension: widens ($\nu < 0$)



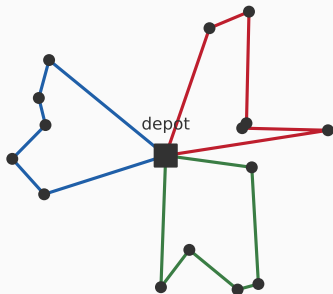
Decision the cell geometry: angles, lengths and thicknesses of the bars

Objective Poisson's ratio ν (the more negative, the better), the stiffness, or both

Constraints minimum manufacturable thickness; allowable stresses; material volume

Type continuous, nonlinear, often **multiobjective** (ν vs. stiffness). Chs. 4–6. Ongoing research at IDMEC

Logistics — delivery routes



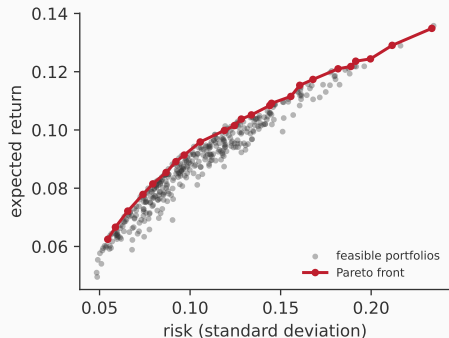
3 vehicles, 15 customers

Decision which vehicle serves each customer and in what order

Objective the total distance (or time, or cost) travelled

Constraints capacity of each vehicle; customers' time windows; maximum shift duration

Type **discrete / combinatorial** — the number of solutions grows like $n!$; core outside the scope of this course; in Ch. 5 only the idea (SA/GA with permutations)



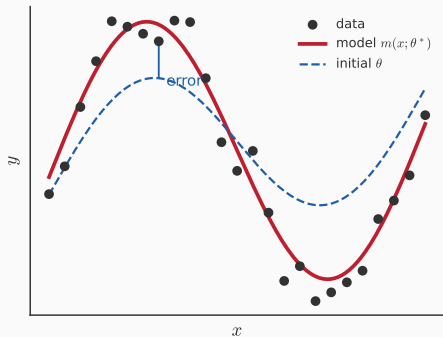
Decision the fraction w_i of capital in each asset

Objective two conflicting criteria: the expected return $\sum w_i \mu_i$ (maximize) and the risk $\sqrt{\mathbf{w}^T \mathbf{C} \mathbf{w}}$ (minimize)

Constraints $\sum w_i = 1$; $w_i \geq 0$ (no short selling); limits per sector

Type **multiobjective**: there is no single solution, there is a Pareto front. Ch. 6 (Markowitz, 1952)

Machine learning — fitting a model

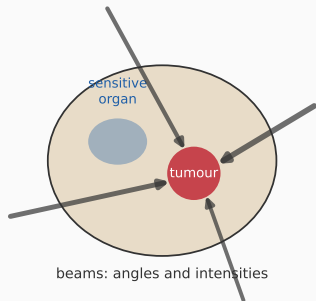


Decision the parameters θ of the model $m(x; \theta)$ — from two to billions

Objective the error on the data, $\sum_k (y_k - m(x_k; \theta))^2$, plus a regularization term

Constraints typically none

Type typically **no explicit constraints** and a huge number of variables; modern training uses gradient-based methods (SGD and variants), whose core idea is that of the steepest descent method of Ch. 3

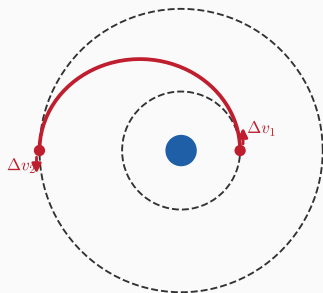


Decision the angle and intensity of each beam

Objective the dose in the tumour (maximize) and the dose in healthy tissue (minimize)

Constraints minimum dose throughout the tumour; maximum dose in each sensitive organ; treatment time

Type continuous, with many constraints, multiobjective. One plan is computed per patient. Chs. 4 and 6



impulses: times, directions and magnitudes

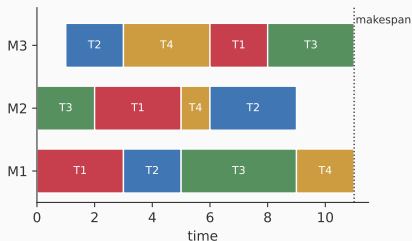
Decision the times, directions and magnitudes of the impulses Δv

Objective the fuel spent, $\sum \|\Delta v_k\|$ (or the travel time)

Constraints the equations of motion; reaching the target orbit; launch windows

Type continuous, nonlinear, with equality constraints (the dynamics). Ch. 4

Production — machine scheduling



Decision the order of the jobs on each machine and the start time of each one

Objective the total time to finish everything (makespan), or the delays relative to the deadlines

Constraints each machine does one job at a time; precedence between jobs; availability

Type **discrete**, combinatorial: heuristics (e.g. GA) and integer programming. Core outside the scope; in Ch. 5 only the idea (SA/GA with permutations)

OPTIMIZATION IN PRACTICE — assigning vehicles to public transport services (Tecmic)

Decision which vehicle (and which driver) runs each service on each day, in a public transport network.

Variables a vehicle $\rightarrow$ service assignment for each service: *binary*, in the thousands.

Objective operating cost (empty kilometres, overtime hours) and workload balance.

Constraints each service covered once; compatibility of timetables and locations; labour rules; depot capacity.

Why it is hard combinatorial: the number of possible assignments grows faster than any power of the number of services; and the problem is solved with real operational data.

Methods assignment/routing models and heuristics; the choice of variables (per service, per block, per shift) changes the difficulty by orders of magnitude.

The same structure — delivery routes (CTT), hospital distribution (Lusíadas Saúde), bicycle redistribution (Gira/EMEL), industrial scheduling (Exide) — appears in student projects of this course with companies. Outside the scope of the continuous methods of Chs. 2–4 and 6 (in Ch. 5, only the idea), but the question is the same: variables, criterion, constraints, cost of evaluating.

The nine cases on the course map

Case	Variables	Constraints	Objectives	Where in the course
Truss	continuous	yes	1	Ch. 4
Wind farm	continuous	yes	1, multimodal	Ch. 5
Auxetic structure	continuous	yes	1 or 2	Chs. 4–6
Delivery routes	discrete	yes	1	Ch. 5 (idea)
Portfolio	continuous	yes	2	Ch. 6
ML model	continuous	no	1	Ch. 3
Radiotherapy	continuous	yes	2	Chs. 4, 6
Orbital trajectory	continuous	yes (equality)	1	Ch. 4
Scheduling	discrete	yes	1	Ch. 5 (idea)

Before choosing a method, classify the problem: that is what determines what works.

And also...

- Inventory and supply management
- Location of warehouses and antennas
- Telecommunication networks
- Power plant dispatch
- Water resources management
- School and crew timetabling
- Design of wings, hulls and profiles
- Industrial process control
- Robotics and motion planning
- Calibration of simulation models
- System identification
- Image and signal processing
- Precision agriculture
- Sport: training and strategy
- Urban planning and traffic
- Advertising and pricing
- Project management (critical path)
- Bioinformatics

In each one: what is decided, what is measured, what limits. Try to identify them.

Summary table — Chapter 1

What information do we have? How much does it cost to obtain? What can we guarantee with it?

Problem	Available information	Cost of evaluating	Guarantee	What made it solvable / hard
fence, can	f and f' explicit	zero	global optimum (1 variable, unimodal)	eliminate a variable and differentiate
production (LP)	linear throughout	zero	global optimum at a vertex	the linear structure
transport network	linear, thousands of variables	zero	global optimum	an LP solver
GALP/ENACOL	model + real data	low, but the model is large	solution of the <i>model</i>	integer decisions, time, uncertainty

KEY TAKEAWAYS — Chapter 1

1. To optimize is to choose the best *feasible* alternative: without well-defined variables, criterion and constraints there is no optimization problem — there is a wish.
2. A model is a simplification; the decision is worth what the model is worth, not the algorithm.
3. Without convexity, a local minimum may not be global; the methods of Ch. 5 try to escape local minima and find globally competitive solutions.
4. Real problems (like the nine cases seen earlier) differ from textbook ones in the *cost* of evaluating f , the lack of derivatives, the discrete variables and the conflicting objectives.
5. The question to ask of any method: what information it uses, how much it costs, what it guarantees.

Next (2.1): a single variable, the building block that Ch. 3 uses in each iteration.

References Full list: *Course bibliography*, at the end of the course PDF.

- Arora, J. S., *Introduction to Optimum Design*, 3rd ed., Academic Press, 2011 — Chs. 2 and 14.
- Markowitz, H., *Portfolio selection*, The Journal of Finance 7(1):77–91, 1952.

Resources: student projects on the Tecmic problem (LMATE and MAI internships, 2022–2026), with simulated annealing and NSGA-II — see 5.2, 6.6 and the *Course bibliography*.